

# Response to the Editor and Reviewers

Submission ID: 20e66410-c472-42dc-ab99-957c24a43f96

Original title: Auditing observability bias in InSAR-derived subsidence exposure estimates with a reproducible geospatial workflow

Revised title: Spatial aggregation alters population-weighted estimates of InSAR observation support

We thank the Editor and Reviewers for their comments. We have revised the estimand and interpretation and added matched-input comparisons of population allocation and InSAR support. For a fixed population model, geographic domain and support field, these comparisons quantify how aggregation changes the summary and identify conditions under which the effect reverses or is small. We have withdrawn the interpretation of the original 3.77-million figure as a verified count of people exposed to subsidence in unobserved locations.

The additional experiments address three questions: whether the aggregation mechanism persists across inputs; how missingness affects estimation error and availability; and how the findings transfer across processing stages, products and regions. Each response below links the relevant result to the reviewer’s question and states its validation scope.

Analyses of irregular blocks, spatial and acquisition-date dependence, training-only quality rules, municipal eligibility and regional transfer address the scope of the revised findings (Supplementary S15, S20–S21, S27, S29–S30). Concentrated missingness leaves substantial allocation unestimated, the added regions have smaller aggregation effects, residential coverage is incomplete, and external GNSS agreement differs between regions.

Location convention: responses identify the numbered main sections and Supplementary sections. The location index at the end gives pages in the accompanying final PDFs. Original reviewer wording is preserved.

## Editor

**Comment.** In particular, the scientific gap and novelty of the study should be more clearly established and supported by a comprehensive and balanced review of the relevant literature. The rationale for the proposed workflow and its methodological choices should be strengthened and better positioned in relation to existing approaches. Further clarification and justification are also required regarding the interpretation and validation of the “hidden exposure” concept, the treatment of observability, the limitations of the GHSL data, and the selection of the InSAR datasets.

**Response.** We have revised the Introduction around established InSAR quality control, published exposure practices and population areal interpolation, followed by the allocation, missingness and transfer questions. We also updated the literature comparison with recent work on nationwide population/building/infrastructure exposure (Xiong et al., 2026), building-level deformation-risk diagnosis (Yang et al., 2026), and uncertainty-aware coastal population-weighted relative sea-level rise (Oelsmann et al., 2026). These studies use broader inventories, target-level data or calibrated vertical-land-motion products; we distinguish their endpoints from our fixed-input allocation audit and do not imply that they validate our LOS observations. The revised gap does not depend on showing that earlier studies generally misused missing support. It asks how replacing a joint fine-scale population-support distribution with coarse summaries changes the reported quantity under fixed inputs, and under which conditions. The covariance decomposition explains the mechanism; the contribution is its quantitative evaluation across matched inputs, scales, origins and application settings, not a new mathematical identity.

The experimental response comprises matched-input scale/origin comparisons, controlled masking with known reference allocations, expanded sampling, alternative population products, a separate connected LiCSBAS experiment with excluded-edge residuals, and a Fresno application using

actual published observation polygons. A further complete-interior analysis shows that the 1-km uniform-allocation difference remains 58.24-58.56% after excluding boundary-crossing cells. This addresses a possible geometric alternative explanation. Internal rechecks and relocated-input checks are documented in Supplementary S7.

The supported finding is that the population-weighted support summary changes substantially when the same population total is redistributed uniformly within coarse cells rather than retained at native model resolution. This allocation-and-aggregation effect is quantified with matched-input tests, controlled references and a published observation-footprint case. It is not a count of verified residents over subsiding ground. GHSL/WorldPop disagreement is treated as population-model sensitivity, and the incomplete Iran branch is excluded from quantitative applications.

The covariance diagnostic in Supplementary S8 agrees with the accounting identity to within  $1.22 \times 10^{-15}$  percentage points. Its pixel-center sensitivity values are kept separate from the direct-geometry estimates. The crossed population-model and pair-selection comparison gives a positive aggregation difference in all 16 cases, with much smaller relative effects for WorldPop than for GHSL. R1.3 provides the numerical results, and Supplementary S18 gives the common-domain design and complete table. Origins represent design sensitivity; pair number and composition change together.

The six further analyses in S15, S20–S21, S27, S29–S30 test these claims against actual irregular reporting geometry, contiguous holes, spatial dependence, training-only quality rules, distinct regions and ancillary sources. These analyses identify specific limits: concentrated missingness leaves much more allocation unestimated than random removal; the added regions have smaller effects; the residential common domain fails the exploratory 90% diagnostic coverage gate; and GNSS agreement is not uniform. The residential gate was introduced before that script produced results; it was not preregistered in protocol v1 or taken from an external standard.

**Changes and evidence.** Main Sections 1–2, 3.1–3.7 and 4; Figures 1–6 and Tables 1–2. Supplementary Figure S20 and Table S29 distinguish the evidence roles. Supplementary S1–S31 retain the complete experiment evidence and the original-estimate audit.

## Code archive and submission files

**Response.** The scientific reproduction package is available at the fixed GitHub release v0.6.2. The reviewer guide provides installation, experiment-specific reproduction and numerical-comparison commands. The scientific baseline from v0.6.0 includes code, processed inputs, reference results and provenance for the six additional analyses, with 21 numerical output comparisons. Release v0.6.2 preserves those numerical results and adds the current submission documents, portable audits and a section-to-output index under `revision_2026/submission_20261003/`. Companion assets provide portable inputs and selected complete provider sources for S15, S20–S21, S27 and S29–S30. The complete ACS files and the two official S25 Census archives remain available at v0.4.1, with checksums. Other provider-controlled inputs and the complete raw processing stacks are obtained through the documented acquisition routes. The earlier DOI 10.5281/zenodo.21444768 identifies v0.3.0 only; a revised DOI deposit remains pending. The clean and numbered manuscripts contain the same current scientific text. The current document snapshot is identified separately from earlier manuscripts retained in the archive.

The additional Bangkok feasibility audit accompanying S17 is supplied separately as `Local_Validation_Audit.zip`, with frozen inputs, all exploratory sampling options and an offline replay command; it does not establish independent accuracy. The product-support check in S3 is supplied as `SBKK_Support_Audit.zip`, including a lossless local filtered-data window, geometry, dates, reference outputs and a standalone replay script. This check reproduces the support statistics and is not a GNSS accuracy comparison.

**Location.** Main Code Availability, cover letter and the versioned reviewer guide.

# Reviewer 1

## R1.1

**Comment (original wording).** The manuscript presents a valuable methodological contribution to the Earth science community by addressing an important but usually overlooked issue »> how InSAR product support masks affect downstream exposure estimates. There are some critical point that must be addressed. 1- The "Hidden Exposure" claim is not validated. The authors should justify such a claim in a scientific way as they address other issues. The manuscript repeatedly emphasizes that it is not validating deformation in hidden cells, yet the central result (3.77 million people in "hidden exposure") is presented as if it carries decision-relevant weight.

**Response.** We agree that the original 3.772492-million statistic did not establish how many people live above subsiding ground in unobserved locations. We withdraw that interpretation and the associated measurement-priority claim. The number sums coarse-cell populations selected by a center-majority rule; neither within-cell residence nor missing deformation is identified. Supplementary S2 retains the original calculation for traceability. The title, definitions, Figure 2, Results and Conclusions now concern population-weighted observation support.

The revised claim is an allocation effect under fixed inputs. Native GHSL allocation gives a mean area-pair support deficit of 3.020 million allocation units; uniform 1-km allocation gives 4.763–4.770 million. These averages are not counts of particular residents. Geometry, conservation, covariance and controlled-mask comparisons check this estimand. Excluding boundary-crossing cells retains a 58.24–58.56% difference. Table 2 now presents the key quantitative checks directly. On the separate matched LiCSBAS grid, the allocation shift is 3.066–3.189 pp for mean input support but only 0.108–0.117 pp for the final mask (S19). We therefore do not treat sampled support as final-product failure.

We also state what can be bounded without inferring missing motion. Under the actual binary final mask, unsupported allocation is 0.2944% of the domain total. At LOS rate  $\leq -5$  mm yr<sup>-1</sup>, observed class allocation is 0.2797% and its whole-domain share lies in [0.2797, 0.5741]%, conditional on the observed rates and population model (S13). Both endpoints are attainable by changing only missing class assignments. The narrow whole-domain interval still permits more than twice the observed rare-class allocation. This missing-mass bound uses a binary mask; the continuous mean support deficit cannot substitute for it. It does not propagate errors in the observed rates or population model.

To assess a local independent comparison, we screened archived CUSV/CUUT GNSS daily series against the 104 Bangkok epochs, using official signed LOS components and the containing InSAR pixel. Exact-date matches number 97/66 and exploratory  $\pm 7$ -day matches 101/77. The stations are only 8.6 m apart and share one pixel; an independent datum common to their global-frame series and the relative InSAR inversion is absent. We report this feasibility result in S17 and supply both spatial samples and date windows in the accompanying `Local_Validation_Audit.zip`. We do not report the frame-mismatched residuals as accuracy. Matched independent Bangkok validation remains unavailable.

The remaining experiments answer separate questions. The 44 excluded edges test internal reconstruction with shared acquisition dates, not external accuracy (S17). Controlled masking tests errors and unestimated allocation against known conditional references (S12–S15). Population benchmarks do not establish household locations (S24–S27). California GNSS compares a different product, with mixed regional agreement and documented frame/model limitations (S30). Supplementary Figure S20, Supplementary Table S29 and the Discussion distinguish these evidence roles. The revised claim is consequently narrower than the submitted physical-exposure claim, and the new experiments test that narrower claim directly.

**Changes and evidence.** Main Sections 2.1, 3.1, 3.5–3.7 and 4.5; Figures 2 and 6; Supplementary Figure S20 and Table S29. Supplementary S2, S12–S15, S17, S19 and S23–S30 distinguish numerical accounting, estimator availability and validation scope.

## R1.2

**Comment (original wording).** P 10: The classification of cells as "majority observable" vs. "not-majority-observable" simplifies a continuous problem. I think binary threshold is unsatisfying the goals and scopes you defined before.

**Response.** We have made support continuous in the primary analysis and report population-, built-surface- and area-weighted distributions over  $[0,1]$ , rather than making the majority split the principal result. We retain the original classification structure as Supplementary Figure S1; its caption states that it is not the revised primary estimand. Figures 3–4 provide the continuous and sampling analyses. The mean deficit equals the integral of the weighted empirical support distribution; we verify this from the exact step distribution. Cutoffs 0.2, 0.3 and 0.4 and unwrapped validity alone remain sensitivity definitions.

The mean does not distinguish spatially persistent missingness from variable pass frequency, so these are not relabeled as the same object. Nor is the 28-pair diagnostic treated as a complete inversion network: it has 54 dates and 26 components. We report expanded 28/56/112/224-pair samples, temporal strata and 100 conditional draws per sample size, while keeping the separate connected processing experiment distinct. The original and nested 28-pair samples differ and are now explicitly identified.

The new training-only quality curve in S21 reports coverage continuously alongside internal reconstruction error instead of reducing quality to a majority label. Shared-acquisition deletion in S20 removes every incident edge for a date and retains the disconnected graph diagnostics; its influence ranges are sensitivity measures, not iid uncertainty intervals.

**Changes and evidence.** Main Sections 2.1, 2.3, 2.6, 3.1–3.4 and 3.7; Figures 3–4; Supplementary Figures S1 and S6. Supplementary S2, S16–S21 retain the continuous, sampling and quality diagnostics.

## R1.3

**Comment (original wording).** P9-10: "GHSL layers are used because they provide consistent public support for population and built-up surface across domains." »> GHSL data limitations are understated and the authors should improve this statements.

**Response.** Population-model sensitivity is now quantified directly. On the common valid domain the  $q=0.3$  deficit is 13.37% for GHSL 2020 and 23.20% for WorldPop 2020. Equalizing the population totals still gives 2.989 versus 5.186 million units, so total-population differences alone do not explain the contrast. We report each model's domain, joint coverage, excluded population and normalized results.

GHSL 2015/2020 epochs are compared separately at fixed 30-arc-second resolution; native-model allocation is compared across reporting scales. Fractional WorldCover summaries preserve an explicit uniformity assumption within GHSL pixels. GHSL and WorldPop remain census/model-based allocations, not local demographic truth. GHSL population and built-surface layers share information, so a built-weighted comparator is not independent validation. Built surface is not occupied floor area; population is not vulnerability, loss or damage. Model differences are not calibrated error bars.

The new crossed experiment tests the aggregation effect itself across population model and pair selection. With the common domain and population total fixed, uniform-minus-native differences are 55.38-55.64% and 51.29-51.51% for GHSL under 28 and 224 pairs, respectively; WorldPop gives 9.03-9.07% and 8.64-8.68%. All 16 model-sample-origin cases have a positive difference, while the magnitude is strongly model dependent. Four origins are sensitivity cases, not independent replicates. Pair number and composition change together. Locations: main Results, "Crossing population model and pair selection" paragraph; Supplementary S18.

S20 now uses spatially separated held-out groups and 0/1/3-km buffers for training-only total recalibration against ACS. Held-out WAPE remains 33.24–33.85% for GHSL and 32.34–32.98% for WorldPop, so normalization does not remove local disagreement. This tests recalibration of

released grids, not their original training. S27 adds independent municipal residential eligibility, with strict/broader semantic rules and an explicit unmapped category. Its joint domain covers only 89.13–89.22% of complete Census population and falls below the exploratory 90% diagnostic gate. The limited-domain changes are reported as allocation sensitivity, not independent household accuracy; source vintage and shared Census lineage remain explicit.

**Changes and evidence.** Main Sections 2.5, 3.2 and 4.5. Supplementary S18, S20 and S22–S27 report population disagreement, spatial recalibration and residential coverage; Supplementary Fig. S16 gives the population-model comparison.

## R1.4

**Comment (original wording).** there are also some minor issues that must be double-checked by the authors. I recommend major revisions with particular attention to clarifying the weak points and the policy-relevant interpretation of the results too.

**Response.** We have checked the scientific quantities, units, dates, sample counts, figure captions, source attribution and cross-references. The revisions distinguish the original 3.772492-million center-majority total, an early revision estimand and the current 3.020-million mean deficit. They also distinguish source-product coverage, sampled pair support and final LiCSBAS quality. The corrected Iran attribution, unconfirmed VLM scale, population-model dependence and temporal mismatches are explicitly retained.

Supplementary Figure S6 reports descriptive ranking stability under actual pair selections. We remove policy claims about a verified population hidden from subsidence assessment and do not propose a validated measurement-priority ranking. The practical recommendation is limited to transparent reporting of the domain, support definition, allocation assumptions, sensitivities and unresolved information. We checked the reported quantities, units, source attribution and document locations.

The tables, source units, periods and control-site exclusions were checked against the source records. The section-to-output index identifies the released tables; the location index below identifies the final document pages. Conflicting archived A137 configuration dates, incomplete municipal coverage and negative GNSS outcomes remain explicit. The public package recomputes the six new analyses and compares all 21 new CSV tables, including per-block, per-tile, replicate and station outputs.

**Changes and evidence.** Main Sections 3–5 and Table 1. Supplementary S1–S3, S12–S15 and S27–S30 retain corrected units, source attribution and limits.

## Reviewer 2

The manuscript "Auditing observability bias in InSAR-derived subsidence exposure estimates with a reproducible geospatial workflow" by Zixuan Liu and Wei Xiong discusses a very important topic regarding InSAR products and how they might translate into misleading exposure estimates. While the topic is relevant and the issues raised by the manuscript are valid, I have major concerns about the structure of the paper.

### R2.1

**Comment (original wording).** 1- The problem formulation gives too little credit to established InSAR practices. It discusses the problem in the Introduction. For example:

"exposure accounting can convert this missingness into a denominator change."

"A processing team may publish a careful product with valid-support masks, while a downstream analyst may treat the remaining raster or point layer as a complete hazard domain."

"but they rarely audit whether non-observed radar support is being treated as absent"

I find the concern relevant and valid. But the literature review does not systematically show that previous InSAR exposure studies have failed to address areas with missing information. The

manuscript claims that "When the resulting deformation map is intersected with population, built-up land or infrastructure layers, analysts often remove cells that are not valid in the radar product before exposure is summarized," without providing evidence from studies that follow this practice. In my opinion, this is the main weakness of the manuscript. If the issue is misuse of correctly documented InSAR products by non-specialist users, this should be stated explicitly and supported by multiple examples of studies where such problems occur, together with references to state-of-the-art studies that have tried to address the issue. In its current form, the manuscript sometimes reads as if the InSAR community has failed to recognize this limitation. However, it is well accepted in the community that InSAR observations are opportunistic by nature. In some cases, InSAR products are accompanied by quantitative information on the validity and quality of the observations. For example, LiCSBAS provides average coherence, the percentage of valid unwrapped pixels, etc. along with velocity and time-series products. The manuscript acknowledges that experienced InSAR users understand and correctly interpret areas where InSAR does not provide information. Therefore, the scientific gap should be demonstrated specifically at the point where InSAR products are used in downstream applications. Without enough evidence from previous studies, it is difficult to determine whether the paper addresses a genuine unresolved research problem.

**Response.** We agree that the original Introduction did not substantiate a general claim that earlier InSAR studies mishandled missing support. We removed those assertions. The revised review recognizes established PS, SBAS, distributed-scatterer, LiCSBAS and MintPy practice, together with published coherence screening, valid-support diagnostics and geodetic comparisons. Our research question no longer depends on demonstrating misuse by earlier authors.

We now locate the question at an explicit downstream operation. Blackwell et al. (2020) assumed uniform population within each community when estimating exposure; Ohenhen et al. (2025) assigned census-block population to subsidence classes using each block's median InSAR-pixel rate. Both also used GNSS comparisons. These examples establish different documented allocation choices, not reporting errors. Geodetic comparisons evaluate the displacement product; they do not by themselves test whether an allocation summary is preserved when fine-scale population-support association is aggregated. Fernandez-Torres et al.'s quality screening and GPS checks, published Bangkok comparisons, and Murray et al.'s stated gaps in motion observations further provide the measurement context (Introduction and S3).

The revised gap has three testable parts. First, how large is the allocation effect when the population model, support field, domain and total are fixed, and does it persist away from boundaries? Second, how does it change across population models, pair selections and input versus final-product support? Third, can low error where an estimator remains available coexist with substantial unestimated allocation? We recognize areal interpolation and dasymetric mapping as established methods and the covariance decomposition as an established identity. The contribution is the matched-input evidence answering these questions, not a new interpolation theory or InSAR processing algorithm.

The complete-interior result retains a 58.24–58.56% primary-case difference. All 16 population-model/pair-selection/origin cases have the same sign, but WorldPop gives much smaller relative effects than GHSL (S18). On the same LiCSBAS grid, the final-mask shift is 0.108–0.117 pp, compared with 3.066–3.189 pp for input support (S19). At 30% artificial removal on irregular Census blocks, prevalence leaves 0.20% of allocation unestimated for random pixels versus 23.01–28.64% for contiguous holes (S15). These results quantify dependence on the inputs and missingness pattern; they do not supply a universal correction factor.

Our median-rule experiment adapts a published classification step to a different fine-allocation target. Undefined units remain undefined, errors use a common-computable domain, and the unmasked baseline records the target difference before missingness (S14–S15). Smaller regional effects and sign reversals are retained. The California product and GNSS comparisons likewise distinguish the provider's physical-validation task from our allocation task (S29–S30). The Introduction, Table 2 and Discussion now connect these conditional results to method choice rather than claiming general superiority or deficiencies in prior practice.

**Changes and evidence.** Main Sections 1, 2.3, 3.2, 3.5–3.7 and 4. Supplementary S3, S6, S14–S15 and S18–S30 provide literature context, method targets and limits.

## R2.2

**Comment (original wording).** 2- The manuscript explains how the proposed workflow works, but it does not adequately explain why this particular method is chosen. State-of-the-art methods and alternative approaches should be discussed and compared with the proposed method. The Materials and Methods section also contains very few references to previous methodological works. As a result, it is difficult to determine how the proposed approach relates to existing methods and why the specific choices made in the workflow are preferable or necessary.

**Response.** The workflow is chosen to isolate an allocation assumption under identical observations. We hold the population model, support field, geographic domain and population total fixed, and preserve extensive quantities during transfer. Under these conditions, differences between summaries can be attributed to allocation. Methods and Supplementary S6 now place these choices within established areal interpolation and dasymetric mapping, citing Eicher and Brewer and Mennis, and compare their inputs, assumptions and evaluation targets.

Each comparator represents a practical choice. Center sampling is inexpensive but can miss subcell and boundary structure. Uniform-area allocation is adequate when the omitted population–support covariance is small relative to the reporting tolerance. Native-model allocation preserves that association when retaining the model’s fine structure is appropriate; it does not validate the model itself. Built weights introduce ancillary assumptions and share information with GHSL. A complete-case overlay describes its observed domain. Imputation, additional sensors and ground observations address missing deformation and need their own reference observations. We therefore do not claim numerical superiority over methods estimating a different quantity.

The comparisons test when these choices matter. Four scales and four origins, direct source-cell intersections and a complete-interior analysis separate allocation from geometry; the primary difference remains 58.24–58.56% after excluding boundary cells. The crossed GHSL/WorldPop and 28/224-pair experiment fixes the common domain and total across all 16 cases; a 25-m integration check changes the result by at most 0.000436 pp (S18). A paired LiCSBAS experiment then changes only the support definition, distinguishing mean input support from the final quality mask (S19). Supplementary S9 relates the covariance bound to a declared reporting tolerance, rather than prescribing one universal resolution.

Controlled masks retain sign reversals, small aggregate-bias cases, local errors and estimator availability. The observed-LOS experiment masks values inside an originally valid domain; actual irregular Census blocks add realistic reporting geometry and contiguous holes. Methods are compared on a common-computable domain, and wholly unestimated blocks remain visible (S12–S15). The actual-mask bounds in S13 make missing class allocation explicit without imputing motion. These designs establish conditional method behavior, not the deformation of genuinely missing pixels.

Population benchmarks address the remaining model dependence. GHSL/WorldPop comparisons, Census-normalized CA-POP allocation and ACS estimates separate source totals, within-unit allocation and reporting-unit disagreement (S22–S26). CA-POP shares Census inputs, and ACS does not establish household locations. Buffered recalibration, municipal eligibility, training-only quality curves and additional regions retain spatial dependence, incomplete coverage and small transfer effects (S20–S21, S27, S29). The California GNSS comparison evaluates a different product and has mixed outcomes (S30). The revised Methods and Discussion use these limits to state when each accounting choice is defensible.

**Changes and evidence.** Main Sections 2.3, 2.5–2.6, 3.2, 3.5 and 3.7. Supplementary S4, S6–S21 and S23–S31 retain the numerical, method, population and processing comparisons.

## R2.3

**Comment (original wording).** 3- The "Iran public InSAR products" is based on relatively old study from 2008. More recent nationwide studies of subsidence in Iran are available, including Haghighi and Motagh, 2024 and Payne et al., 2025. InSAR datasets from both of these studies are publicly available. The manuscript should explain why an older dataset was selected instead of these newer products. I think the use of the 2008 dataset does not invalidate the workflow, but the it should be justified.

**Response.** We have removed the exploratory Iran branch from the quantitative applications and corrected its source attribution. We therefore no longer present the manuscript as having selected an older Iranian dataset for a completed case study, and we do not claim a new/old product comparison.

The checksum audit identifies the archived rate and mask as the 2014–2020 product accompanying Haghighi and Motagh (2024), rather than a 2008 nationwide rate product. The seasonal-amplitude archive does not match the companion checksum and is unused. Supplementary S1 and S3 distinguish that national annual-rate product from Payne et al.'s 2014–2022 selected-region vertical/east–west velocities and uncertainties. Those summary layers have different targets and are not pair-specific valid-unwrapped/coherence masks. Source interferograms remain separately available; we do not claim the newer products are inaccessible or inferior. The quantitative applications and conclusions use the declared Chao Phraya and Fresno inputs, with the separately documented California transfer cases in S27, S29–S30.

The two additional ARIA regions and their GNSS comparison are documented in S29–S30. These quantitative applications use the archived California products described there.

**Changes and evidence.** Main Section 4.5, Earlier estimates and source attribution; Supplementary S1 and S3. California inputs and comparisons are separately documented in S27 and S29–S30.

## Final document location index

The page numbers below refer to the accompanying clean manuscript and Supplementary Information. The numbered manuscript uses the same section labels. Main Table 2 and Supplementary Table S29 summarize the analyses and their scope.

| Comment | Main sections and PDF pages                                               | Supplementary sections and PDF pages                                                |
|---------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Editor  | 1 (p. 2), 2 (p. 4), 3.2 (p. 10), 4 (p. 17)                                | S1 (p. 2), S6 (p. 7), S7 (p. 8), S14 (p. 12), S30 (p. 29), S31 (p. 32)              |
| R1.1    | 2.1 (p. 4), 3.1 (p. 9), 3.5 (p. 13), 3.7 (p. 15), 4.5 (p. 19)             | S2 (p. 3), S12 (p. 11), S13 (p. 12), S15 (p. 13), S17 (p. 17), S30 (p. 29)          |
| R1.2    | 2.1 (p. 4), 2.3 (p. 5), 3.2 (p. 10), 3.3 (p. 11), 3.7 (p. 15)             | S2 (p. 3), S16 (p. 14), S20 (p. 20), S21 (p. 21)                                    |
| R1.3    | 2.5 (p. 8), 3.2 (p. 10), 4.5 (p. 19)                                      | S18 (p. 18), S20 (p. 20), S22 (p. 22), S25 (p. 24), S26 (p. 25), S27 (p. 25)        |
| R1.4    | 3 (p. 9), 4 (p. 17), 5 (p. 20)                                            | S1 (p. 2), S2 (p. 3), S3 (p. 4), S14 (p. 12), S28 (p. 27), S30 (p. 29)              |
| R2.1    | 1 (p. 2), 3.2 (p. 10), 3.5 (p. 13), 3.6 (p. 14), 4 (p. 17)                | S3 (p. 4), S6 (p. 7), S14 (p. 12), S15 (p. 13), S18 (p. 18), S29 (p. 28)            |
| R2.2    | 2.3 (p. 5), 2.5 (p. 8), 2.6 (p. 8), 3.2 (p. 10), 3.5 (p. 13), 3.7 (p. 15) | S4 (p. 6), S6 (p. 7), S7 (p. 8), S14 (p. 12), S18 (p. 18), S19 (p. 20), S31 (p. 32) |
| R2.3    | 4.5, Earlier estimates and source attribution (p. 20)                     | S1 (p. 2), S3 (p. 4), S27 (p. 25), S29 (p. 28), S30 (p. 29)                         |

Supplementary Sections S1–S31 provide the primary and California experiment results. S13 reports the bounds and S17 describes the local GNSS feasibility analysis supplied separately in `Local_Validation_Audit.zip`; this analysis does not establish independent Bangkok accuracy. The scientific code release is identified in Code Availability.